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OBJECTIVE/SUMMARY

A proficient mechanical engineer with a strong mathematics background, mechanical/product design with practical machinery experience. A wide range of experience with 3-D CAD, and drafting detailed drawings. A mechanical engineer with a breadth of skills, experience, and talent. Creative, mature, technically sound, capable engineer interacting closely with manufacturing, quality, sales, and marketing departments. Proficient at designing/carrying out experiments and finalizing technical reports. Experienced in designing, testing, documenting, validating, and implementing commercial products for market.

EDUCATION

BS Mechanical Engineering with Minor in **Mathematics**
University of Delaware: Newark, DE
EIT Certification

EXPERIENCE

Maxon Precision Motors Taunton, MA
New Product Development Manufacturing Engineer 1/2024 – Current

- Maxon Precision Motors designs and manufactures world class brushed and brushless DC motors. Individual motors and integrated subsystems are part of maxon's offering. Many manufacturing techniques are utilized from rapid prototyping to high quantity tooling.

Amphenol Alden Products Brockton, MA
Senior Design Engineer 6/2022 – 6/2023

- Amphenol Alden Products designs and manufactures electrical connectors for the medical device industry. Their products are relied upon for the monitoring and treatment of patients in healthcare settings.

ThermoFisher Scientific Franklin, MA
Mechanical Engineer III 2/2021 – 6/2022

- The Franklin, MA ThermoFisher Scientific site produces air quality monitoring equipment for use in environments where contaminants in the air are a concern. ThermoFisher Scientific has been integral in supporting the fight against Covid-19.

IDEX Health & Science Middleboro, MA
<u>Senior Design Engineer</u> 4/2017 – 11/2020

- IDEX Health & Science develops and manufactures components for instruments used in IVD/Bio applications. Specifically, the Middleboro location is associated with fluidics, and precision displacement pumps.

Vacuum Process Technology Plymouth, MA
<u>Senior Mechanical Engineer</u> 10/2016 – 3/2017

- Vacuum Process Technology designs and manufactures vacuum chambers for electron beam, and thermal deposition processes.

Testing Machines, INC. New Castle, DE
<u>Senior Mechanical Engineer</u> 2/2012 – 9/2016

- Testing Machines, Inc. designs and manufactures precision material testing instruments for the adhesive, corrugated, film, foil, ink, paper, tissue, plastic, pulp, rubber, and textile markets.

SCHOTT DiamondView Armor Products Boothwyn, PA
<u>Design Engineer</u> 10/2009 – 10/2011

- Schott DiamondView Armor Products designs and manufactures transparent armor, and transparent armor related housing components for military and private applications.

AMS Filling Systems, Inc. Honey Brook, PA
<u>Lead Design Engineer</u> 9/2008 – 8/2009

- AMS Filling Systems, Inc. designs and manufactures solid and liquid filling equipment and feeders for the pharmaceutical, food and beverage, cosmetic, and chemical markets.

ThermoFisher Scientific (Formerly NanoDrop Technologies) Wilmington, DE
<u>Product Development Engineer</u> 5/2006 – 7/2008

- The NanoDrop division of ThermoFisher Scientific designs and manufactures novel spectrophotometric instruments for the cell biology, genomics, immunology, sequencing, and bioengineering markets.

PROFESSIONAL SKILLS

- **New Product Design and Development**
  - Development process and life cycle
  - Accelerated Life Testing (ALT), Highly accelerated life testing (HALT)
  - **Materials** utilized in design include steel, stainless steel, aluminum, copper, brass, titanium, tungsten, sheet metal (parts and weldments), polycarbonate, delrin, polystyrene, polypropylene, polyethylene, urethane, ABS, PMMA, PVC, PEEK, nylon, teflon, UHMW, garolite, borosilicate glass, sapphire, quartz, alumina, Buna N, EPDM, fluorosilicone, neoprene, santoprene, viton, fiberglass,

acrylic and epoxy adhesives, optical cement, solder, lacquers, dessicant, and wood.

    - **Finishes** utilized in design include wet paint, powdercoat, electrodeposited and electroless nickel plating, zinc phosphate plating, electrodeposited zinc plating, chrome plating, gold plating, galvanizing, passivating, electropolishing, black oxide, anodizing, alodining, hydrophobic coating, hydrophilic coating, vapor polishing, vapor deposition, and plasma treatment.
- **Computer Aided Design (CAD)**
    - Solidworks, AutoCAD, Draftsight
    - 3D-modeling, 2D detailed drawings, simulation, motion studies, parametric studies, and photorealistic rendering
- **Geometric Dimensioning and Tolerancing (GD&T)**
    - ASME Y14.5
    - Tolerance stack analysis
- **Design for Manufacture / Assembly (DFMA)**
    - Facilitate and lead DFMA activities
    - Boothroyd-Dewhurst approach for DFMA
    - DFMA cost estimates for quoting purposes
- **Machinery / Tools**
    - CNC vertical mills, lathes, drill presses, band saws, sand blasters, arbor presses, belt sanders, air compressors, tumblers, vibration tables, water jet, plasma, and laser cutting machines, autoclaves, sheet metal brakes, rollers, grinders, soldering irons and guns, brazing and cutting torches, spectrometers, light sources, haze meters, XYZ robots, 3D printers, calipers, micrometers, dial indicators, depth gauges, gauge pins, gauge blocks, flow meters, multimeters, and oscilloscopes
- **Sensors and DAQ**
    - Measurement Computing DAQ boards and software, Texas Instruments DAQ boards, and LabView software programming
    - Force and pressure transducers, thermocouples, accelerometers, strain gauges, encoders, mechanical and optical limit switches, reed switches, flexible resistive force and pressure sensors, and capacitance sensors
- **Microsoft Office**
    - Microsoft Word, Excel, PowerPoint, Project, Teams, Visio, Outlook, and Access
- **Finite Element Analysis (FEA)**
    - Solidworks Simulation, ANSYS
    - Stress, strain, displacement, thermal, and flow analysis
- **Project Management**
    - Business requirements, functional requirements, incoming inspection, cost analysis, price point margins, timeline scheduling, gantt charts, flow diagrams, risk management, quality management, resource management, feasibility studies, training, review, and transition plans

- **Electrical Components**
    - Stepper and servo motors, linear actuators, push buttons, touch screens, stepper and servo motor drive boards, capacitors, resistors, inductors, transformers, logic gates, breadboards, relays, DAQ modules, power sources and entry modules, terminals, breakers and fuses, solenoids, cameras, LCD screens, and LEDs
- **Engineering Change Control**
    - ISO 9001
    - Paper process
    - Solidworks EPDM
- **Interpretation of Standard Specifications**
    - ISO, ASTM, TAPPI, DIN, MIL, FMVSS
    - CE conformance risk analysis, and associated technical data packages
- **Optical Components**
    - Lenses, prisms, fiber optic cables, lasers, cameras, optical encoders, optical stages, optical benches, filters, optically clear materials, optical cement, integrating spheres
- **Pneumatic/Fluidic Components**
    - Air cylinders, regulators, filters, dryers, push-to-connect and barbed fittings, air springs, manifolds, solenoid valves, exhaust valves, relief valves, and pressure boosters
- **Enterprise Resource Planning (ERP)**
    - Syteline, JD Edwards, IFS, and Infor XA
    - BOM generation and structuring, material and labor routings, vendor cross-referencing, and cost input
- **Computer Aided Machining (CAM)**
    - BobCAD
    - G-Code for CNC Vertical Mill
- **Programming**
    - C
    - proprietary X-Y-Z laboratory robot language
    - LabView

PATENTS

- US 10661268 (Grant): Pipette Tip System, Device and Method of Use
- US 9638608 (Grant): Pipette Tip System, Device and Method of Use
- US 9533427 (Grant): Corrugated Sample Cutter
- US 7623225 (Grant): Photometer with Modular Light Emitter
- CN 206613510 U (Grant): Pipette Tip System, Device and Method of Use
- US 10946373B2 (Grant): Pipette Tip System, Device and Method of Use
- US 17210709 (Application): Precision Volumetric Pump with a Bellows Hermetic Seal
- US 63/352,355 (Application): Systems and Methods for Providing a High Efficiency Solar-Powered Mechanical Generator for Generating Electrical Power